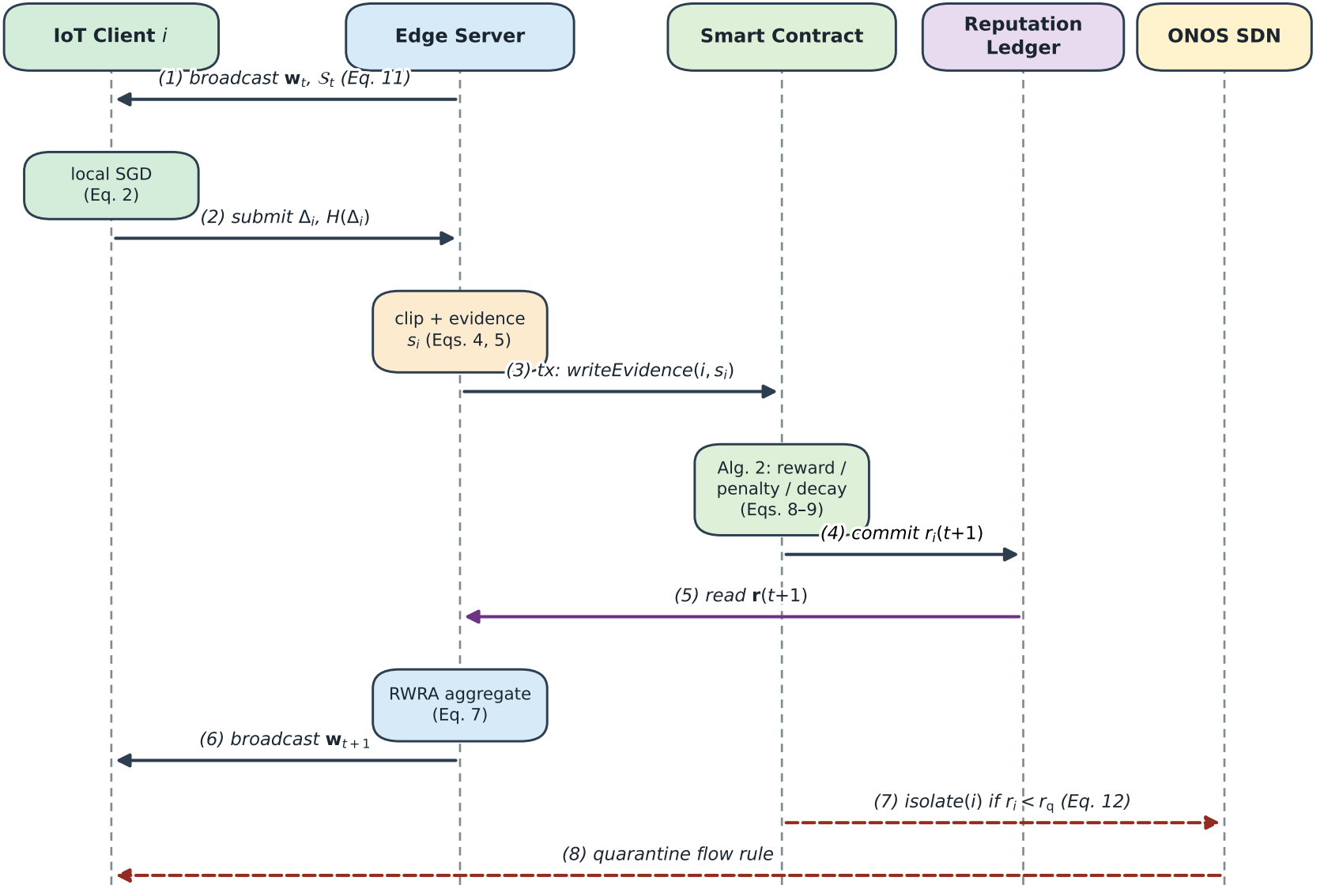

IoT Client i
Edge Server
Smart Contract
Reputation Ledger
ONOS SDN
(1) broadcast $\mathbf{w}_t$, $\mathcal{S}_t$ (Eq. 11)
local SGD (Eq. 2)
(2) submit Δ_i, $H(\Delta_i)$
clip + evidence s_i (Eqs. 4, 5)
(3) tx: writeEvidence(i, s_i)
Alg. 2: reward / penalty / decay (Eqs. 8–9)
(4) commit $r_i(t+1)$
(5) read $\mathbf{r}(t+1)$
RWRA aggregate (Eq. 7)
(6) broadcast $\mathbf{w}_{t+1}$
(7) isolate(i) if $r_i < r_q$ (Eq. 12)
(8) quarantine flow rule